Lecture 19: Poisson Process

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1 The Poisson Process

Question

Crashes at a busy intersection occur at *random times*. How should we model
- **when** crashes happen, and
- **how many** crashes happen in a given window of time?

Reasonable modeling assumptions

- Crashes are equally likely at any moment (no rush hour, in this simple model).
- What happens in one window of time tells us nothing about a disjoint window.
- In a very short window, we see at most one crash.

This is the **Bernoulli process** with time chopped into many tiny slots.

But, how tiny does it need to be so that in reality each window has at most one crash? Here, a continuous model would be more natural!

1.1 Poisson process: definition

Definition

An arrival process with **arrival rate** $\lambda > 0$ (arrivals per unit time). Let $P(k, \tau)$ be the probability of exactly k arrivals in an interval of length τ. The process satisfies:

1. **Time homogeneity:** $P(k, \tau)$ is the same for every interval of length τ.
2. **Independence:** the numbers of arrivals in disjoint intervals are independent.
3. **Small intervals** (length $\delta \to 0$):

$$P(0, \delta) = 1 - \lambda\delta + o(\delta), \qquad P(1, \delta) = \lambda\delta + o(\delta), \qquad P(k \geq 2, \delta) = o(\delta).$$

Reading the definition

$\lambda\delta$ plays the role of the Bernoulli success probability p in a tiny slot, and $o(\delta)$ means "negligible compared to δ".

1.2 Number of Arrivals in an Interval

Derivation

Chop $[0, \tau]$ into $n = \tau/\delta$ slots. Each slot is an independent Bernoulli trial with success probability

$$p = \lambda\delta = \frac{\lambda\tau}{n},$$

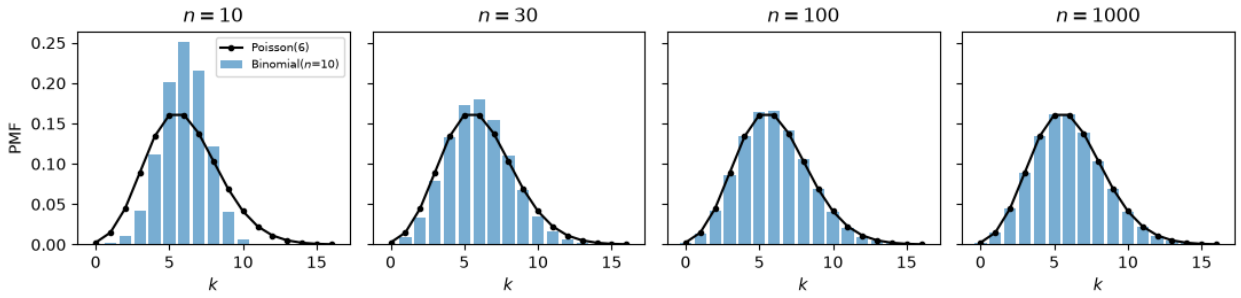
so the number of arrivals $N_\tau \sim \text{Binomial}(n, p)$.

Let $n \to \infty$ with $np = \lambda\tau$ fixed

$$\binom{n}{k} p^k (1-p)^{n-k} \longrightarrow \frac{(\lambda\tau)^k}{k!} e^{-\lambda\tau}.$$

We saw this limit in Lecture 3; here is where it comes from.

```
mu = 6.0      # lambda * tau
k = np.arange(0, 17)
fig, axs = plt.subplots(1, 4, figsize=(11, 2.8), sharey=True)
for ax, n in zip(axs, [10, 30, 100, 1000]):
    ax.bar(k, stats.binom.pmf(k, n, mu / n), color="C0", alpha=0.6, label=f"Binomial($n$={n})")
    ax.plot(k, stats.poisson.pmf(k, mu), "ko-", ms=3, label="Poisson(6)")
    ax.set_title(f"$n={n}$"); ax.set_xlabel("$k$")
axs[0].set_ylabel("PMF"); axs[0].legend(fontsize=7)
plt.tight_layout(); plt.show()
```



1.3 Poisson PMF, mean, and variance

Poisson PMF for the number of arrivals in time τ

$$P(N_\tau = k) = P(k, \tau) = \frac{(\lambda\tau)^k e^{-\lambda\tau}}{k!}, \qquad k = 0, 1, 2, \ldots$$

Mean and variance

$$E[N_\tau] = \lambda\tau, \qquad \mathrm{Var}(N_\tau) = \lambda\tau.$$

These match the Binomial values $np = \lambda\tau$ and $np(1-p) \to \lambda\tau$ in the limit $p \to 0$.

1.4 Time of the First Arrival

Tail probability

Let T_1 be the time of the first arrival. No arrival by time t means zero arrivals in $[0, t]$:

$$P(T_1 > t) = P(0, t) = e^{-\lambda t}, \qquad t \geq 0.$$

CDF → PDF

$$F_{T_1}(t) = 1 - e^{-\lambda t} \quad \Longrightarrow \quad f_{T_1}(t) = \frac{d}{dt} F_{T_1}(t) = \lambda e^{-\lambda t}, \quad t \geq 0.$$

Conclusion

$T_1 \sim$ Exponential(λ), with

$$E[T_1] = \frac{1}{\lambda}, \qquad \mathrm{Var}(T_1) = \frac{1}{\lambda^2}.$$

2 Summary: Bernoulli vs. Poisson

2.1 Two descriptions of arrivals

	Bernoulli process	**Poisson process**
Times of arrival	discrete: $i = 1, 2, \ldots$	continuous: $t \geq 0$

	Bernoulli process	**Poisson process**
Arrival rate	p per trial	λ per unit time
PMF of # of arrivals	Binomial(n, p)	Poisson$(\lambda\tau)$
Interarrival time	Geometric(p)	Exponential(λ)
Mean interarrival time	$1/p$	$1/\lambda$
Time of kth arrival	Pascal(k, p)	Erlang(k, λ)

Take-away

Poisson is the continuous-time limit of Bernoulli: **Binomial** $\to$ **Poisson** and **Geometric** $\to$ **Exponential**.

2.1.1 Example: emails

Example

Emails arrive as a Poisson process with rate $\lambda = 5$ per hour.

(a) What is the probability of no emails in the next 30 minutes?
(b) What is the probability of exactly 3 emails in the next hour?
(c) What is the expected wait until the next email?

Solution

(a) $\lambda\tau = 5(0.5) = 2.5$, so $P(N = 0) = e^{-2.5} \approx 0.082$.
(b) $\lambda\tau = 5$, so $P(N = 3) = \dfrac{5^3 e^{-5}}{3!} \approx 0.140$.
(c) $E[T_1] = 1/\lambda = 1/5$ hour $= 12$ minutes.

2.2 Independence and Memorylessness

Fact

For any time t, the arrivals after t form a new Poisson process with the same rate λ, **independent of everything that happened before** t.

Memorylessness of the exponential

$$P(T_1 > t + s \mid T_1 > t) = \frac{e^{-\lambda(t+s)}}{e^{-\lambda t}} = e^{-\lambda s} = P(T_1 > s).$$

Having already waited t does not change how much longer we wait.

2.3 Interarrival Times

Definition

Let T_k be the time of the kth arrival, with $T_0 = 0$. The **interarrival times** are

$$Y_k = T_k - T_{k-1}, \qquad k = 1, 2, \ldots$$

Alternative description of the Poisson process

$Y_1, Y_2, \ldots$ are **i.i.d. Exponential**(λ).
Conversely, drawing i.i.d. exponential gaps and placing arrivals at their running sums produces a Poisson process with rate λ.

Why

By the fresh-start property, the process restarts independently after each arrival, so every gap has the same law as T_1.

2.4 The kth arrival time

Definition

$$T_k = Y_1 + Y_2 + \cdots + Y_k,$$

a sum of k i.i.d. Exponential(λ) interarrival times.

Properties (Erlang distribution)

$$f_{T_k}(y) = \frac{\lambda^k y^{k-1} e^{-\lambda y}}{(k-1)!}, \quad y \geq 0, \qquad E[T_k] = \frac{k}{\lambda}, \qquad \mathrm{Var}(T_k) = \frac{k}{\lambda^2}.$$

Derivation sketch

For a small δ: $f_{T_k}(y)\,\delta \approx P(k-1 \text{ arrivals in } [0,y]) \cdot \lambda\delta = \dfrac{(\lambda y)^{k-1} e^{-\lambda y}}{(k-1)!}\lambda\delta$.
Mean and variance follow from the sum of k i.i.d. exponentials.

2.4.1 Example: IRS hotline

Example

Calls arrive at the IRS hotline as a Poisson process with rate $\lambda = 0.5$ per minute.
(a) What are the mean and standard deviation of the time of the 4th call?
(b) What is the probability that the 3rd call arrives after minute 4?

Solution

(a) $E[T_4] = 4/0.5 = 8$ minutes and $\mathrm{SD}(T_4) = \sqrt{4}/0.5 = 4$ minutes.
(b) $T_3 > 4$ exactly when at most 2 calls arrive in $[0, 4]$. With $\lambda\tau = 2$,

$$P(T_3 > 4) = P(N_4 \le 2) = e^{-2}\left(1 + 2 + \tfrac{2^2}{2}\right) = 5e^{-2} \approx 0.677.$$

2.4.2 Exercise

Exercise

Customers arrive at a shop as a Poisson process with rate $\lambda = 6$ per hour.
(a) What is the probability that exactly 2 customers arrive in the next 20 minutes?
(b) What is the expected time until the 3rd customer arrives?
(c) No one has arrived in the last 10 minutes. What is the probability that no one arrives in the next 10 minutes?

Solution

(a) $\lambda\tau = 6 \cdot \frac{1}{3} = 2$, so $P = \dfrac{2^2 e^{-2}}{2!} = 2e^{-2} \approx 0.271$.
(b) $E[T_3] = 3/6 = 0.5$ hours $= 30$ minutes.
(c) By memorylessness, $e^{-6\cdot(1/6)} = e^{-1} \approx 0.368$; the past 10 minutes are irrelevant.

2.5 Splitting

Splitting

Each arrival of a Poisson(λ) process is independently labeled **type A** with probability p and type B otherwise.

Result

Type A and type B arrivals are **independent Poisson processes** with rates

$$\lambda_A = p\lambda, \qquad \lambda_B = (1-p)\lambda.$$

Intuition

In a tiny slot, $P(\text{type A arrival}) = \lambda\delta \cdot p$, and slots remain independent.

2.6 Merging

Merging

Combine two **independent** Poisson processes with rates λ_1 and λ_2 into a single stream.

Result

The merged process is Poisson with rate $\lambda_1 + \lambda_2$, and each arrival independently comes from process 1 with probability

$$\frac{\lambda_1}{\lambda_1 + \lambda_2}.$$

Why

In a tiny slot, $P(\text{arrival from either}) \approx (\lambda_1 + \lambda_2)\delta$. Equivalently, the first arrival time is the minimum of two independent exponentials, which is Exponential$(\lambda_1 + \lambda_2)$.

2.6.1 Example: mail packages at the post office

Example

Packages are mailed as a Poisson process with rate 10 per hour. Each package is domestic with probability 0.7 and international with probability 0.3, independently.

(a) What is the probability of no international packages in 30 minutes?

(b) Given that 4 packages arrived in an hour, what is the probability that exactly 1 is international?

Solution

By splitting, domestic packages are Poisson with rate 7 per hour and international packages are Poisson with rate 3 per hour, independent of each other.

(a) $e^{-3(0.5)} = e^{-1.5} \approx 0.223$.

(b) Each of the 4 packages is international independently with probability 0.3, so $\binom{4}{1}(0.3)(0.7)^3 \approx 0.412$.

2.7 Sums of a Random Number of RVs

Setup

Let $Y = X_1 + \cdots + X_N$, where N is a nonnegative integer-valued random variable and the X_i are i.i.d. and independent of N.

Summary

	Formula
Mean	$E[Y] = E[N]\,E[X]$
Variance	$\mathrm{Var}(Y) = E[N]\,\mathrm{Var}(X) + (E[X])^2\,\mathrm{Var}(N)$
Transform	$M_Y(s) = M_N(\ln M_X(s))$

Where it comes from

Condition on N: $E[Y \mid N] = N\,E[X]$ and $\mathrm{Var}(Y \mid N) = N\,\mathrm{Var}(X)$, then apply the law of iterated expectation and the law of total variance from Lecture 10.

2.8 Random Incidence Paradox

Example

Buses arrive as a Poisson process with rate λ, so the average gap between buses is $1/\lambda$. You arrive at a random time. How long is the gap you land in?

Looking forward and backward

- Time until the next bus: Exponential(λ) with mean $1/\lambda$ (fresh start).
- Time since the last bus: also Exponential(λ) with mean $1/\lambda$ (run the process backward in time).

The paradox

The gap containing your arrival has mean about $2/\lambda$, **twice** the average gap.

Resolution

Random times are more likely to land in **long** gaps. This is length-biased sampling, so the gap you observe is not a typical one.

2.8.1 Example

Example

Gaps between buses are 5 minutes or 15 minutes, each with probability $\frac{1}{2}$, so the average gap is 10 minutes. What is the average gap seen by a random arriver?

Solution

A random arrival lands in a 5-minute gap with probability $\frac{5}{5+15} = 0.25$ and in a 15-minute gap with probability 0.75:

$$E[\text{observed gap}] = 0.25(5) + 0.75(15) = 12.5 \text{ minutes} > 10.$$

In general, $E[\text{observed gap}] = \dfrac{E[Y^2]}{E[Y]}$, which for Exponential($\lambda$) gaps is $\dfrac{2/\lambda^2}{1/\lambda} = \dfrac{2}{\lambda}$.